

TOBB University of Economics and Technology

Department of Electrical and Electronics Engineering

ELE 224L – Electronic Circuits Laboratory

Experiment 3 – Preliminary Work: MOSFET Amplifiers

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1 INTRODUCTION

1.1 Objective of the Experiment

The objective of this experiment is to learn the operating principle of the MOSFET as an amplifying element in the common-source amplifier and as a matched current source in the current mirror. The DC operating point (Q point), the small-signal gain, and the input and output resistances of a common-source amplifier are calculated for the cases with and without a source bypass capacitor, and the current mirror is analysed for a range of input voltages, including the voltage at which it stops working.

1.2 Benefits of the Experiment

The experiment shows that an amplifier analysis consists of two separate steps (a DC bias analysis and a small-signal AC analysis), that a bypass capacitor changes the gain by a large factor, and that the capacitor values must be checked against the test frequency before a mid-band formula is used. It also shows how a current mirror copies a reference current and where this copying fails. These skills are required for the BJT amplifiers of the next experiments.

2 CALCULATIONS AND SIMULATIONS

2.1 Common-Source Amplifier Circuit

The amplifier circuit is shown in Figure 2.1.1. The values $V_{DD}=12$ V, $R_1=47$ kΩ, $R_2=33$ kΩ , $R_D=15$ kΩ, $R_S=10$ kΩ, $R_L=10$ kΩ, $C_{C1}=22$ nF, $C_{C2}=2.2$ nF and $C_S=22$ nF are used. The BS170 is modelled with the square-law equation $I_D=(K_N/2)(V_{GS}-V_{TH})^2$ with $K_N=72$ mA/V² and $V_{TH}=2.0$ V, and the channel-length modulation is neglected.

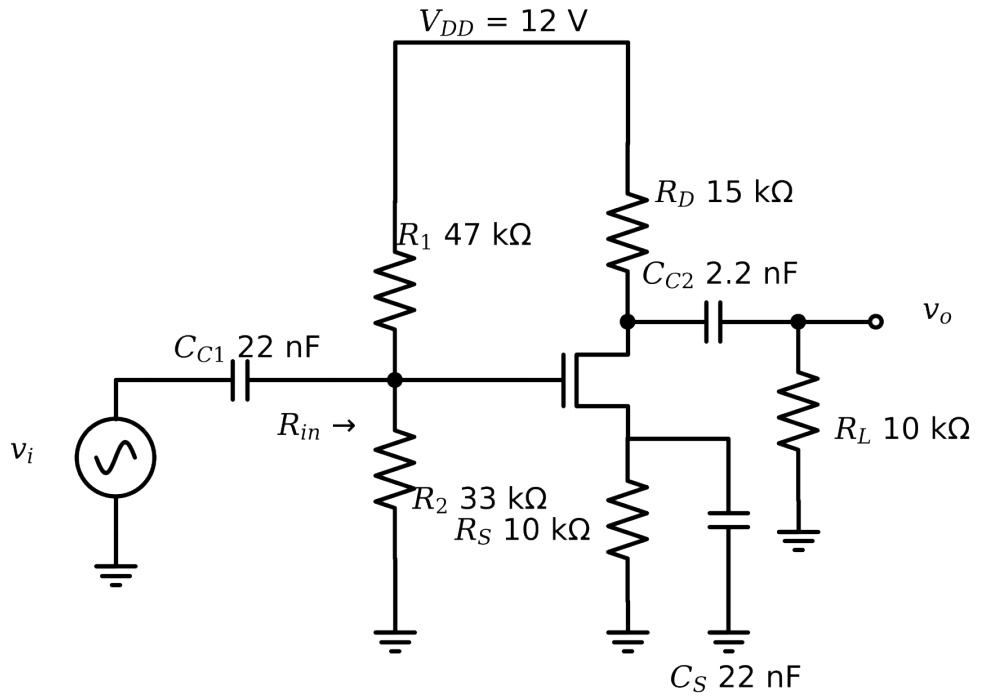


Figure 2.1.1 Common-source amplifier circuit analysed in the experiment.

DC analysis. All capacitors are open circuits for DC. The gate voltage is given by the divider, (2.1). The gate current is zero, so $V_{GS}=V_G-I_D R_S$. Substituting this into the square law and writing $V_{OV}=V_{GS}-V_{TH}$ gives the quadratic equation (2.2).

$$V_G=V_{DD}\frac{R_2}{R_1+R_2}=12\cdot\frac{33}{80}=4.95\ \text{V} \tag{2.1}$$

$$\frac{K_N}{2}R_S V_{OV}^2+V_{OV}-\left(V_G-V_{TH}\right)=0\Rightarrow 360\,V_{OV}^2+V_{OV}-2.95=0 \tag{2.2}$$

The positive root is $V_{OV}=0.08914$ V, so $V_{GSQ}=2.0+0.08914=2.089$ V, $I_{DQ}=(72/2)(0.08914)^2=0.2861$ mA, and $V_{DSQ}=V_{DD}-I_{DQ}\left(R_D+R_S\right)=12-0.2861\times 25=4.848$ V. Since $V_{DSQ}>V_{OV}$, the transistor is in saturation, which verifies the assumption. The transconductance is $g_m=K_N V_{OV}=6.418$ mS.

Small-signal analysis. For mid-band frequencies, the coupling capacitors are short circuits and the supply is an AC ground. With C_S present, the source is at AC ground and the gain is given by (2.3). Without C_S, the source resistor produces local feedback, and the gain is given by (2.4). In both cases, $R_{in}=R_1\parallel R_2$ because the gate draws no current, and $R_{out}=R_D$ because the output resistance of the transistor is infinite in the model (the load R_L is not included in R_{out}).

$$A_V = -g_m\left(R_D \parallel R_L\right) = -6.418\,\text{mS} \times 6\,\text{k}\,\Omega = -38.51 \tag{2.3}$$

$$A_V = -\frac{g_m\left(R_D \parallel R_L\right)}{1+g_m R_S} = \frac{-38.51}{1+6.418\times 10} = -0.591 \tag{2.4}$$

$$R_{in} = R_1 \parallel R_2 = 19.39\,\text{k}\,\Omega,\; R_{out} = R_D = 15\,\text{k}\,\Omega \tag{2.5}$$

Table 2.1.1 Calculated Q point, gain and resistances of the common-source amplifier

	Circuit with C_S	**Circuit without C_S**
V_{GSQ} (V)	2.089	2.089
V_{DSQ} (V)	4.848	4.848
I_{DQ} (mA)	0.2861	0.2861
$A_V = V_o/V_i$ (mid-band, C_S ideal short)	−38.51	−0.5908
R_{in} (kΩ)	19.39	19.39
R_{out} (kΩ)	15.00	15.00
$\left\|A_V\right\|$ at 10 kHz (phasor analysis)	7.689	0.5671

The mid-band formulas assume that C_S is a perfect short circuit. This assumption was checked for the test frequency of 10 kHz. At this frequency, the impedance of C_S is $1/\left(2\pi f C_S\right) = 723\,\Omega$, which is not small compared with $1/g_m = 156\,\Omega$; the parallel combination of R_S and C_S is 722 Ω in magnitude, and the gain is reduced by the factor $\left|1+g_m Z_S\right|$. A complete phasor analysis of the circuit (including C_{C1}, C_{C2} and C_S) gives $\left|A_V\right| = 7.69$ with C_S and 0.567 without C_S at 10 kHz, as listed in the last row of Table 2.1.1. For $\left|A_V\right|$ at 10 kHz to reach 90% of the mid-band value, C_S would have to be about 0.28 µF.

SPICE. The circuit was built with the BS170 model. The Q point obtained from the DC analysis is compared with the calculation in Table 2.1.2. A sinusoidal input of 100 mV amplitude and 10 kHz was applied, and the time-domain graphs are shown in Figure 2.1.2 (with C_S) and Figure 2.1.3 (without C_S); more than two periods are visible. The transient amplitude ratio was measured in steady state. The frequency response obtained with an AC analysis is shown in Figure 2.1.4.

Table 2.1.2 Comparison of the calculation and SPICE for the common-source amplifier (Error = (SPICE − calculation)/calculation)

Quantity	Calculation	SPICE	Error
V_{GSQ} (V)	2.089	2.168	+3.8%
V_{DSQ} (V)	4.848	5.044	+4.0%
I_{DQ} (mA)	0.2861	0.2782	-2.7%
$\lvert A_V \rvert$ at 10 kHz, with C_S	7.689	7.748	+0.8%
$\lvert A_V \rvert$ at 10 kHz, without C_S	0.5671	0.5690	+0.3%
Mid-band $\lvert A_V \rvert$, with C_S (SPICE at 1 MHz)	38.51	48.97	+27.2%
Mid-band $\lvert A_V \rvert$, without C_S (SPICE at 1 MHz)	0.5908	0.5927	+0.3%
R_{in} at 10 kHz (kΩ)	19.39	19.40	+0.07%
R_{out} (kΩ)	15.00	15.00	-0.00%

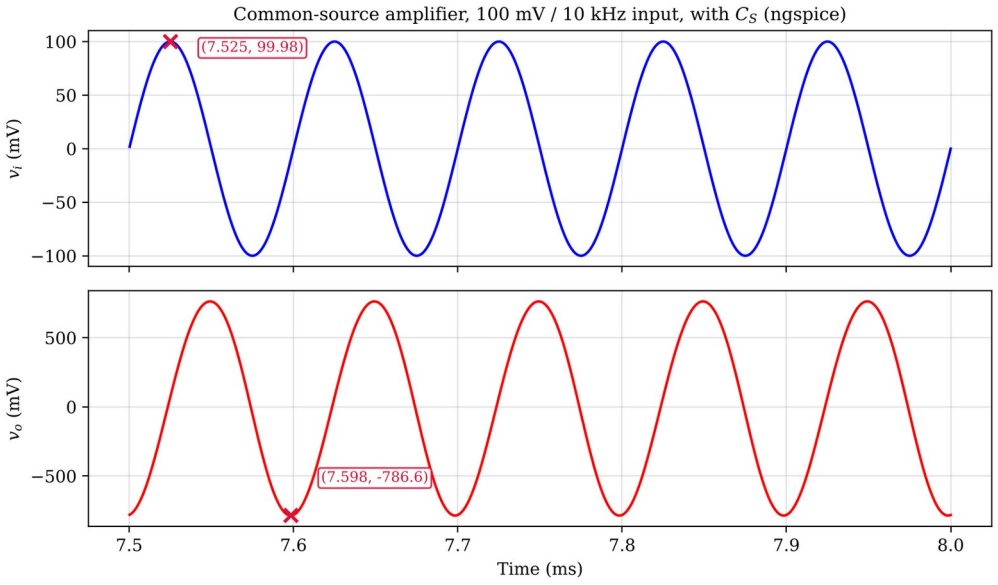


Figure 2.1.2 Input and output voltages of the common-source amplifier with C_S (100 mV, 10 kHz; ngspice).

Table 2.1.3 Coordinates of the cursors in Figure 2.1.2

Cursor	x coordinate	y coordinate
X1	= 7.525	v_i = 99.98 mV
X2	Time = 7.598 ms	v_o = −786.6 mV

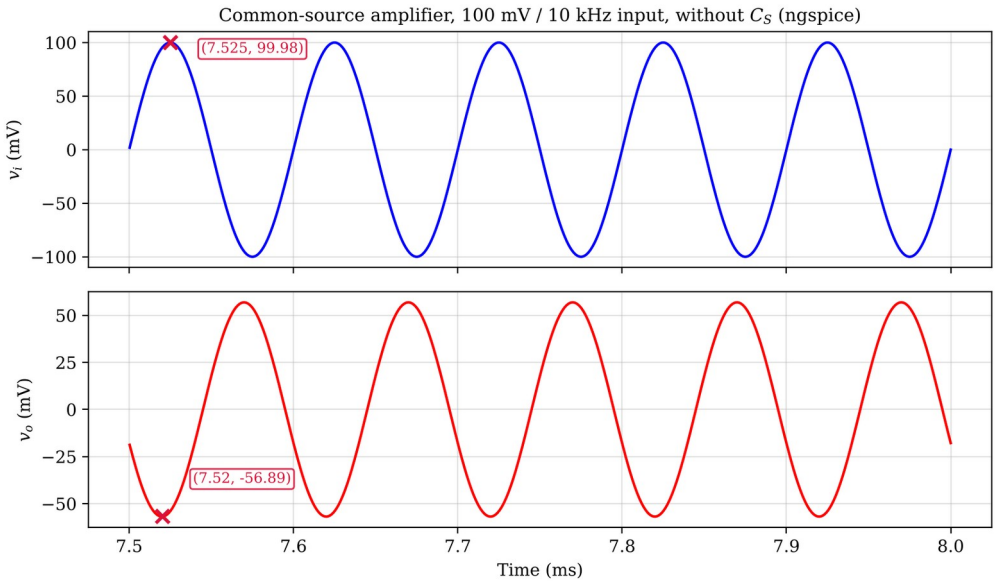


Figure 2.1.3 Input and output voltages of the common-source amplifier without C_S (100 mV, 10 kHz; ngspice).

Table 2.1.4 Coordinates of the cursors in Figure 2.1.3

Cursor	x coordinate	y coordinate
X1	= 7.525	v_i = 99.98 mV
X2	Time = 7.520 ms	v_o = −56.89 mV

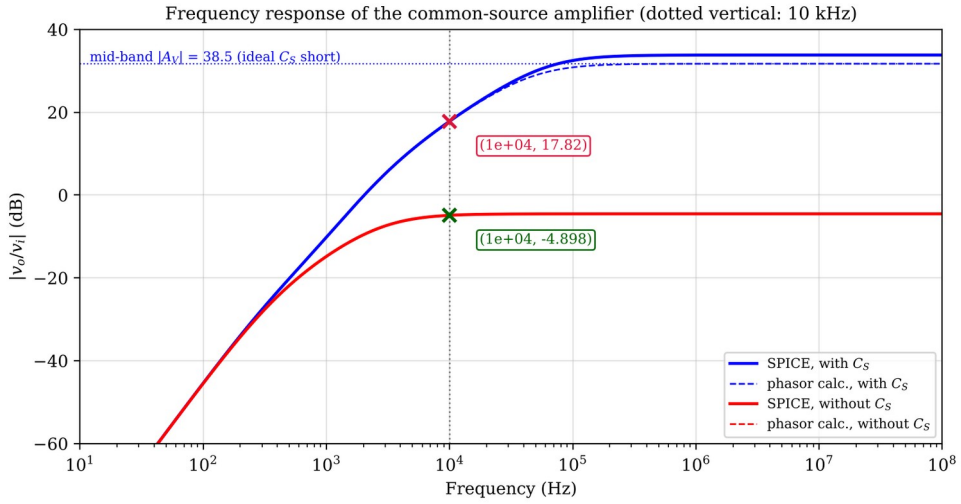


Figure 2.1.4 Frequency response of the amplifier: SPICE (solid) and the phasor calculation (dashed).

Table 2.1.5 Coordinates of the cursors in Figure 2.1.4

Cursor	x coordinate	y coordinate
X1	Frequency = 1.000e+04 Hz	$\lvert v_o/v_i \rvert$ = 17.82 dB
X2	Frequency = 1.000e+04 Hz	$\lvert v_o/v_i \rvert$ = −4.898 dB

The simulation was repeated in LTspice; the waveform window with the cursors is shown in Figure 2.1.5, and the cursor readings are compared with the ngspice values at the same x in Table 2.1.6. The agreement shows that the result does not depend on the simulator.

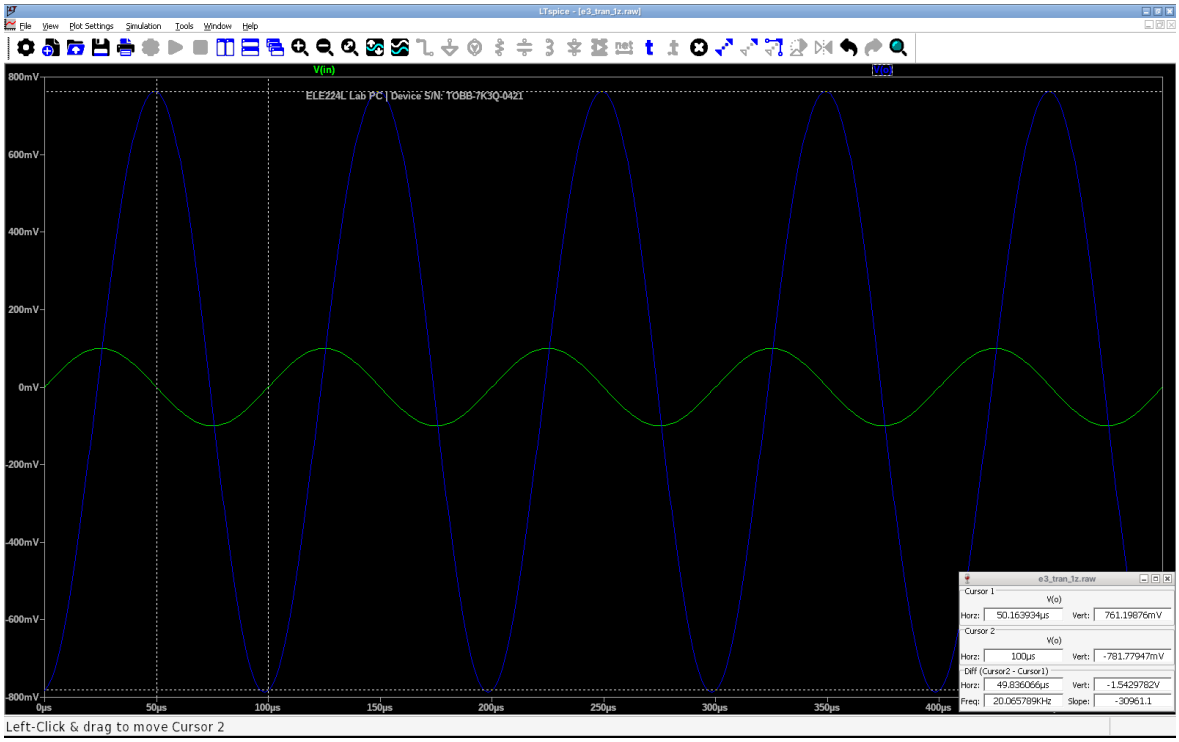


Figure 2.1.5 LTspice transient result of the common-source amplifier with the bypass capacitor C_S (V_{in}: green, V_O: blue); the time axis starts at 7 ms (data saved from 7 ms to 7.5 ms) (device serial number: TOBB-7K3Q-0421)

Table 2.1.6 Cursor coordinates read in Figure 2.1.5 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	t = 7 ms + 50.16 µs	V_O	761.2 mV	761.39 mV	-0.025%
2	t = 7 ms + 100 µs	V_O	−781.78 mV	−781.86 mV	+0.010%

The LTspice result for this case is shown in Figure 2.1.6 and the cursor comparison is given in Table 2.1.7.

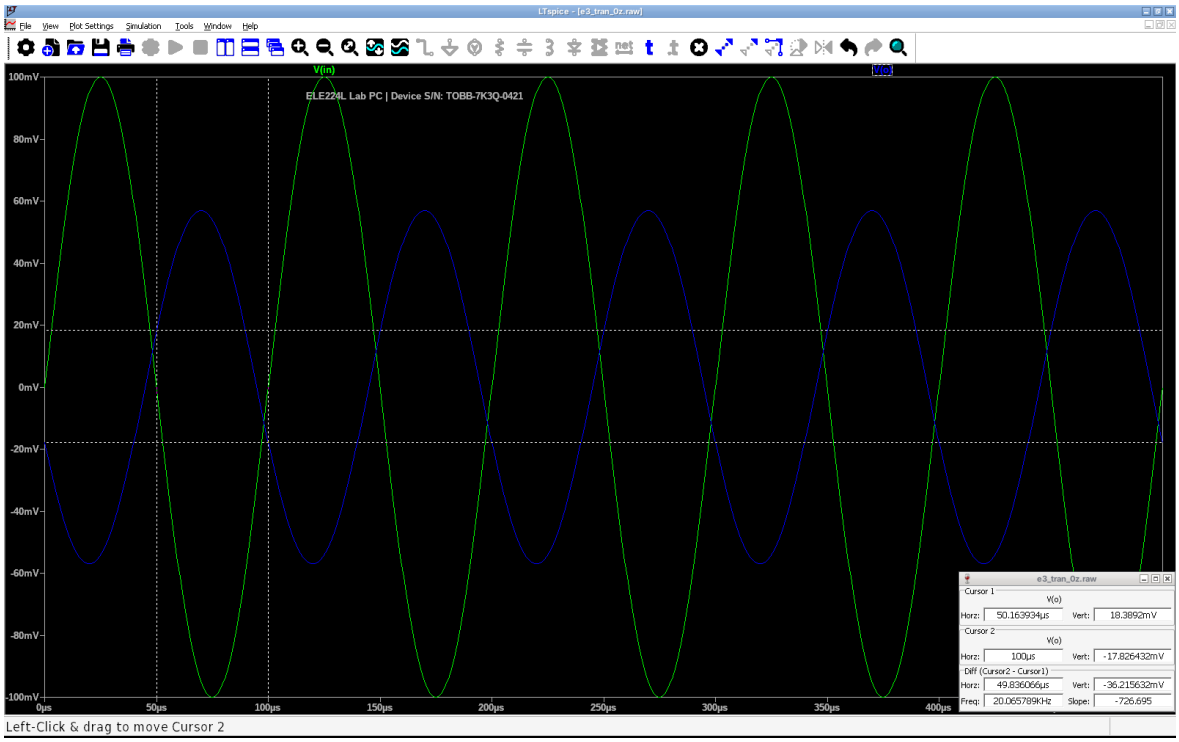


Figure 2.1.6 LTspice transient result of the common-source amplifier without the bypass capacitor (V_{in}: green, V_O: blue); the time axis starts at 7 ms (device serial number: TOBB-7K3Q-0421)

Table 2.1.7 Cursor coordinates read in Figure 2.1.6 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	t = 7 ms + 50.16 µs	V_O	18.389 mV	18.38 mV	+0.050%
2	t = 7 ms + 100 µs	V_O	−17.826 mV	−17.828 mV	+0.011%

The LTspice result for this case is shown in Figure 2.1.7 and the cursor comparison is given in Table 2.1.8.

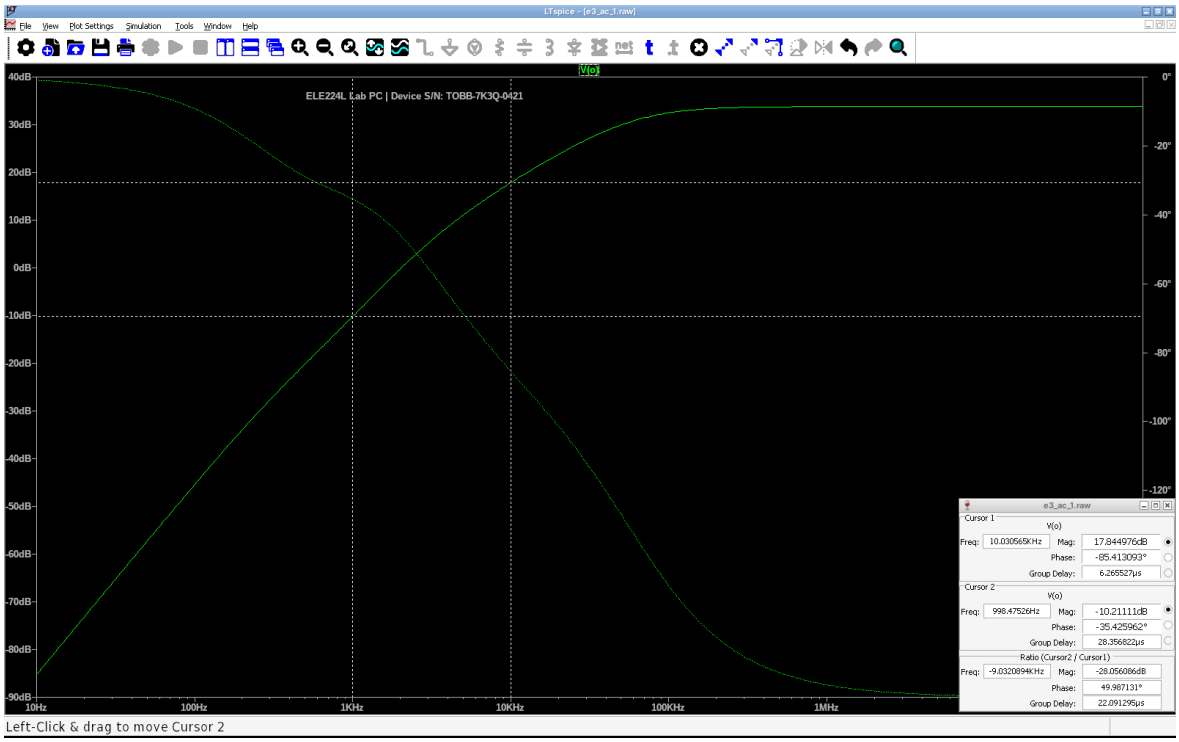


Figure 2.1.7 LTspice AC analysis of the common-source amplifier with C_S (magnitude and phase of V_O for $V_i = 1$ V) (device serial number: TOBB-7K3Q-0421)

Table 2.1.8 Cursor coordinates read in Figure 2.1.7 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	f = 10.0306 kHz	$\lvert V_O \rvert$	17.845 dB	17.845 dB	+0.000 dB
1	f = 10.0306 kHz	$\angle V_O$	−85.413 °	−85.411 °	-0.002 °
2	f = 998.48 Hz	$\lvert V_O \rvert$	−10.211 dB	−10.211 dB	+0.000 dB
2	f = 998.48 Hz	$\angle V_O$	−35.426 °	−35.426 °	-0.000 °

The LTspice result for this case is shown in Figure 2.1.8 and the cursor comparison is given in Table 2.1.9.

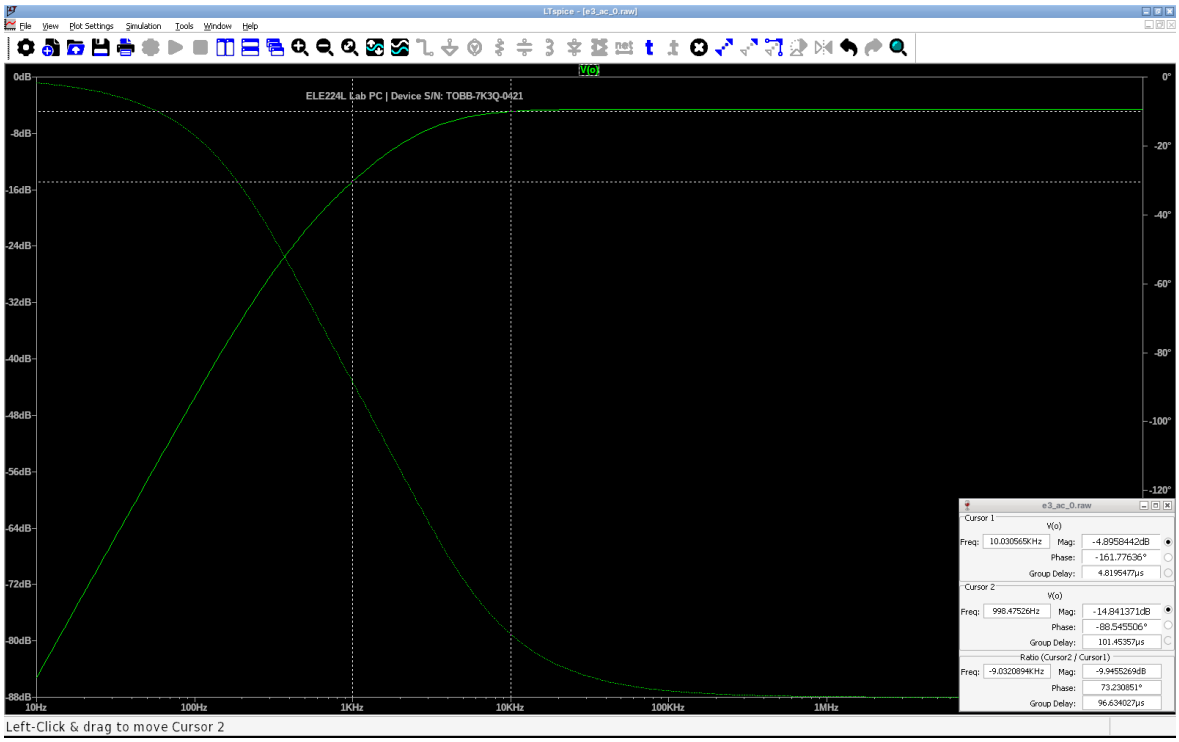


Figure 2.1.8 LTspice AC analysis of the common-source amplifier without C_S (device serial number: TOBB-7K3Q-0421)

Table 2.1.9 Cursor coordinates read in Figure 2.1.8 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	f = 10.0306 kHz	$\lvert V_O \rvert$	−4.8958 dB	−4.8959 dB	+0.000 dB
1	f = 10.0306 kHz	$\angle V_O$	−161.78 °	−161.78 °	-0.001 °
2	f = 998.48 Hz	$\lvert V_O \rvert$	−14.841 dB	−14.842 dB	+0.000 dB
2	f = 998.48 Hz	$\angle V_O$	−88.546 °	−88.544 °	-0.001 °

Interpretation. The DC values of SPICE differ from the calculation by less than 5%, since the SPICE model has a larger transconductance parameter ($KP \cdot W/L = 123.3$ mA/V² against $K_N = 72$ mA/V²), a slightly higher threshold (2.1 V against 2.0 V), and series resistances R_D and R_S of 1.4 and 1.6 Ω. At 10 kHz, the measured gain with C_S is 7.75, which is far below the mid-band value 38.5 of the calculation, but it agrees with the phasor calculation (7.69) to 0.8%; the reason is that at this frequency the gain is determined by the impedance of C_S, $\lvert A_V \rvert \approx R_L' / \lvert Z_S \rvert$, and not by g_m. The mid-band gain of SPICE (49.0) is 27% larger than the calculated mid-band gain since the SPICE transistor has a higher g_m at the same current, and this difference is visible only above about 100 kHz in Figure 2.1.4. Without C_S, the gain is almost independent of g_m because $g_m R_S \gg 1$, $A_V \approx -R_L'/R_S = -0.6$, and the SPICE result agrees with the calculation within 0.3%.

2.2 Current Mirror Circuit

The current mirror is shown in Figure 2.2.1. The transistor M1 is diode connected (its gate and drain are tied), so it is always in saturation, and V_{GS} is set by the reference current I_{ref}. Since M2 has the same V_{GS} and is identical to M1, it carries the same current as long as it stays in saturation, i.e., as long as $V_{DS2} = V_{DD} - I_{out} R_2 \geq V_{OV}$. The reference current is found from the loop equation (2.6), which is solved for I_{ref} by iteration (or as a quadratic in V_{OV}).

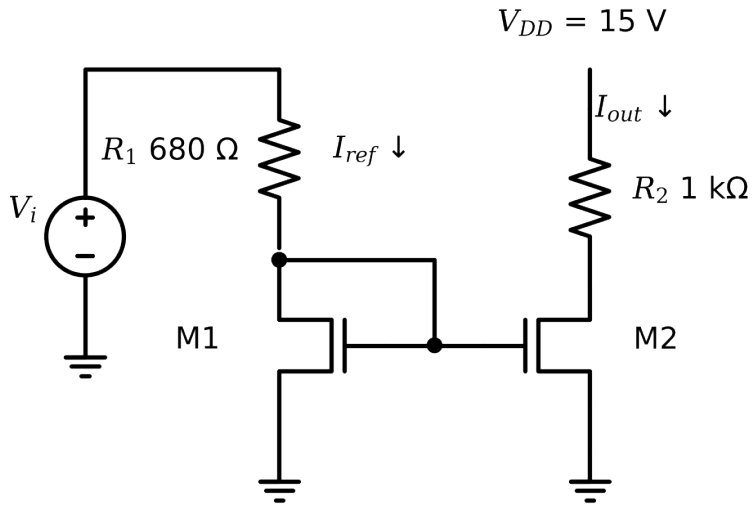


Figure 2.2.1 Current mirror circuit analysed in the experiment.

$$V_i = R_1 I_{ref} + V_{TH} + \sqrt{\frac{2 I_{ref}}{K_N}} \tag{2.6}$$

For example, for $V_i = 3$ V, $V_i - V_{TH} = 1$ V gives the quadratic $24.48 V_{OV}^2 + V_{OV} - 1 = 0$ (with V_{OV} in V and $R_1 = 0.68$ kΩ), and $V_{OV} = 0.182$ V, $I_{ref} = 36 \cdot 0.182^2 = 1.20$ mA. Then $V_{DS2} = 15 - 1.20 \times 1 = 13.8$ V $> V_{OV}$, so M2 is in saturation and $I_{out} = I_{ref} = 1.20$ mA. For $V_i = 1$ V (below V_{TH}), M1 is off and both currents are zero. For $V_i = 15$ V, I_{ref} would be 18.08 mA and M2 would require $V_{DS2} = 15 - 18.1 < 0$, which is impossible; M2 therefore leaves saturation, enters the triode region, and the solution of the triode current equation together with $V_{DS2} = 15 - I_{out} R_2$ gives $I_{out} = 14.60$ mA. The hand-calculated and simulated values are given in Table 2.2.1.

Table 2.2.1 Reference and output currents of the current mirror: calculation and SPICE

V_i (V)	I_{ref} calc. (mA)	I_{out} calc. (mA)	I_{ref} SPICE (mA)	I_{out} SPICE (mA)	M2 region (calc.)
1.00	0	0	≈ 0	≈ 0	cut-off
3.00	1.202	1.202	1.122	1.122	saturation
5.00	3.926	3.926	3.886	3.886	saturation
8.00	8.125	8.125	8.124	8.124	saturation
10.0	10.95	10.95	10.97	10.97	saturation
12.0	13.80	13.80	13.83	13.83	saturation
15.0	18.08	14.60	18.13	14.66	triode

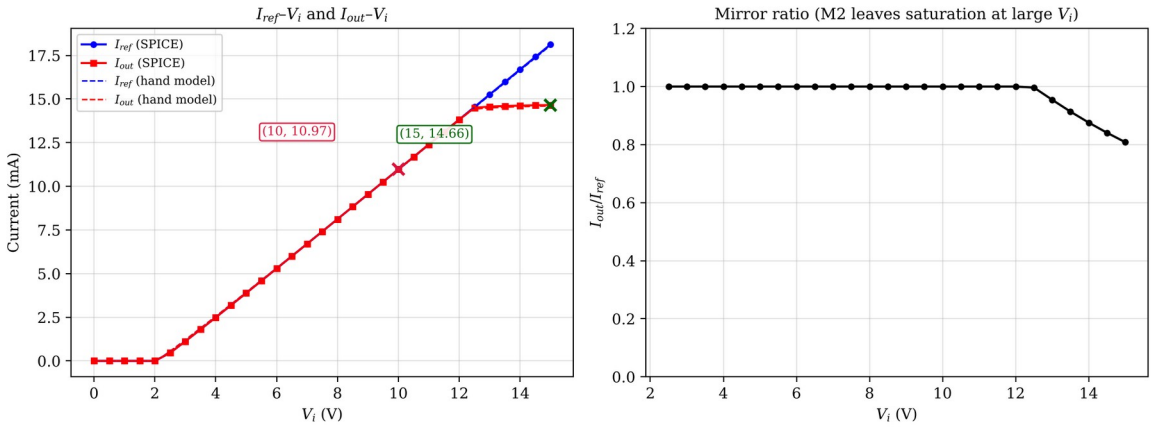


Figure 2.2.2 I_{ref}–V_i and I_{out}–V_i graphs of the current mirror (V_i on the x-axis) with the mirror ratio (ngspice).

Table 2.2.2 Coordinates of the cursors in Figure 2.2.2

Cursor	x coordinate	y coordinate
X1	V_i = 10.00 V	Current = 10.97 mA
X2	V_i = 15.00 V	Current = 14.66 mA

The LTspice result for this case is shown in Figure 2.2.3 and the cursor comparison is given in Table 2.2.3.

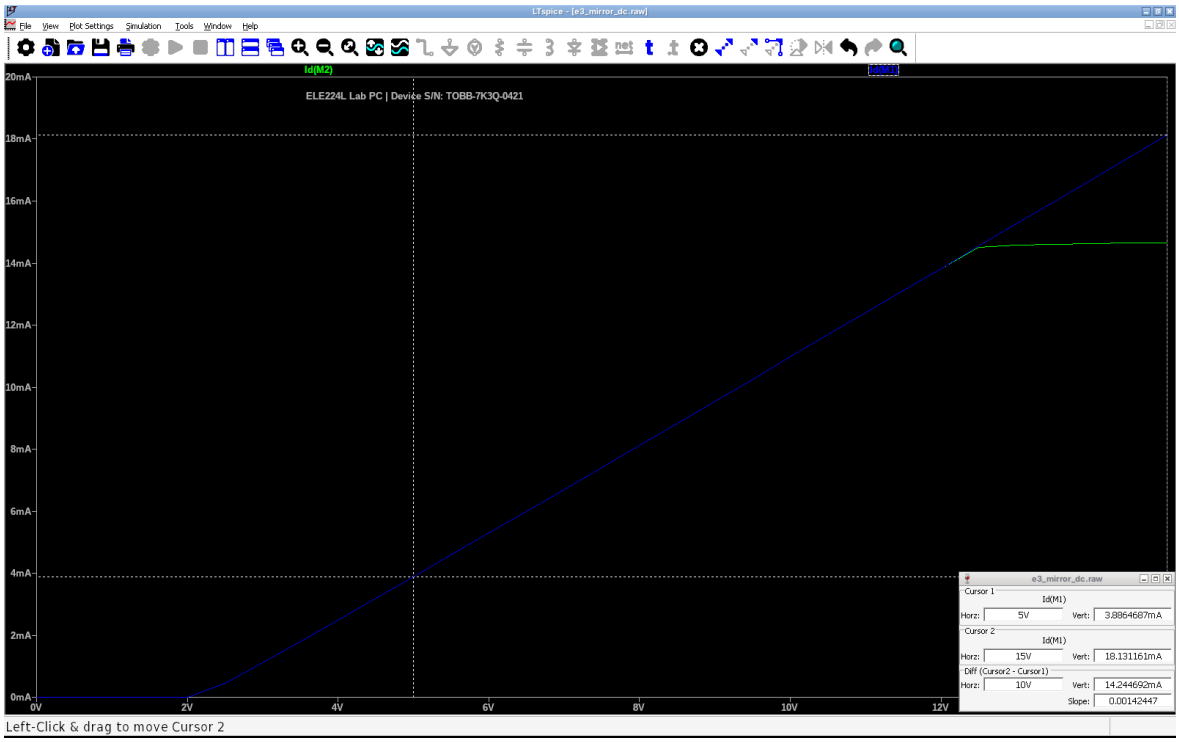


Figure 2.2.3 LTspice DC sweep of the current mirror: reference-branch current $I_D(M1)$ (blue) and output current $I_D(M2)$ (green) versus V_i; the cursors read $I_D(M1)$ (device serial number: TOBB-7K3Q-0421)

Table 2.2.3 Cursor coordinates read in Figure 2.2.3 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	V_i = 5 V	$I_D(M1)$	3.8865 mA	3.8869 mA	-0.010%
2	V_i = 15 V	$I_D(M1)$	18.131 mA	18.131 mA	+0.000%

The two currents are equal (within 0.1%) up to $V_i = 12$ V. The SPICE and calculated reference currents agree within 1.0% for $V_i \geq 5$ V, whereas at 3 V the SPICE current is 6.6% lower than the calculated one; this larger difference is caused by the different transconductance and threshold of the SPICE model when V_{OV} is small. For $V_i = 15$ V, I_{out} stays at about 14.7 mA while I_{ref} continues to rise to 18.1 mA, because M2 enters the triode region (the calculation predicts this for $V_i > 12.4$ V; in the 0.5 V steps of the SPICE sweep, the ratio I_{out}/I_{ref} first falls below 0.99 at 13.0 V). The mirror therefore works as long as the load drop $I_{out}R_2$ leaves at least V_{OV} across M2, i.e., $V_{DD} - I_{out}R_2 \geq V_{OV}$. In saturation, the equality of the currents also relies on the neglect of channel-length modulation, which is zero in the BS170 model.

3 CONCLUSIONS

The calculated Q point of the common-source amplifier ($V_{GSQ} = 2.09$ V, $I_{DQ} = 0.286$ mA, $V_{DSQ} = 4.85$ V) agrees with SPICE within about 4%, with the differences caused by the model

parameters. The most important result is that the mid-band gain formulas cannot be applied at 10 kHz with the given capacitor values: the bypass capacitor of 22 nF is too small, and the measured gain with C_S is 7.75 instead of 38.5. A bypass capacitor of at least about 0.28 µF would be required for this test frequency, or the test frequency should be raised above about 100 kHz. Without the bypass capacitor, the gain is small (−0.591) but stable, since it depends only on the resistor ratio. The analysis also gave $R_{in} = 19.39$ kΩ (set by the bias divider) and $R_{out} = 15$ kΩ (set by R_D); the SPICE values agree within 0.1%.

In the current mirror, the output current follows the reference current until the output transistor loses saturation at about 12–13 V; the experiment taught the limit of the mirror (compliance voltage), the use of a diode-connected transistor as a bias generator, and the difference between an ideal copy and the real behaviour. The unexpected results above were explained by checking the capacitor impedances and the model parameters, which is a skill required in the remaining experiments.

4 REFERENCES

1. Sedra, A. S., & Smith, K. C. (2014). Microelectronic circuits (7th ed.). New York: Oxford University Press.
2. ON Semiconductor. (2017). BS170 small signal MOSFET 500 mA, 60 volts [Data sheet]. Retrieved October 11, 2026, from https://www.onsemi.com/download/data-sheet/pdf/bs170-d.pdf
3. ngspice. (n.d.). ngspice user manual. Retrieved October 11, 2026, from https://ngspice.sourceforge.io/docs.html

5 APPENDICES

5.1 Appendix A: SPICE Netlists

Netlist A.1 Common-source amplifier with C_S (transient)

```
VDD vdd 0 DC 12
Vi in 0 DC 0 AC 1 SIN(0 0.1 10k)
Cc1 in g 22n
R1 vdd g 47k
R2 g 0 33k
M1 d g s 0 BS170 W=100u L=100u
RD vdd d 15k
RS s 0 10k
CS s 0 22n
Cc2 d o 2.2n
RL o 0 10k

.model BS170 NMOS (VTO=2.1 RS=1.572 RD=1.436 KP=.1233 IS=5p)
```

```
.tran 2u 8m 0 1u
.end
```

Netlist A.2 Current mirror (DC sweep)

```
VDD vdd 0 DC 15
Vi vi 0 DC 0
R1 vi d1 680
M1 d1 d1 0 0 BS170 W=100u L=100u
M2 d2 d1 0 0 BS170 W=100u L=100u
R2 vdd d2 1k
.model BS170 NMOS (VTO=2.1 RS=1.572 RD=1.436 KP=.1233 IS=5p)

.dc Vi 0 15 0.5
.end
```